

Fixed Point Iteration Part-2

Theorem 2.3

- (i) If $g \in C[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$, then g has at least one fixed point in $[a, b]$.
- (ii) If, in addition, $g'(x)$ exists on (a, b) and a positive constant $k < 1$ exists with

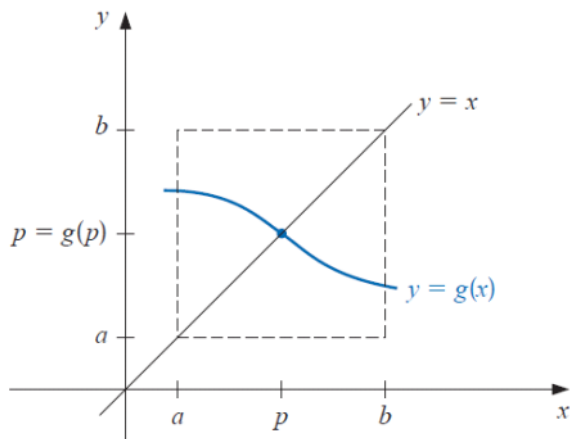
$$|g'(x)| \leq k, \quad \text{for all } x \in (a, b),$$

then there is exactly one fixed point in $[a, b]$. (See Figure 2.4.) ■

Theorem 1.2 (Intermediate Value Theorem). Assume that $f \in C[a, b]$ and L is any number between $f(a)$ and $f(b)$. Then there exists a number c , with $c \in (a, b)$, such that $f(c) = L$.

Theorem 1.6 (Mean Value Theorem). Assume that $f \in C[a, b]$ and that $f'(x)$ exists for all $x \in (a, b)$. Then there exists a number c , with $c \in (a, b)$, such that

$$(11) \quad f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Proof

- (i) If $g(a) = a$ or $g(b) = b$, then g has a fixed point at an endpoint. If not, then $g(a) > a$ and $g(b) < b$. The function $h(x) = g(x) - x$ is continuous on $[a, b]$, with

$$h(a) = g(a) - a > 0 \quad \text{and} \quad h(b) = g(b) - b < 0.$$

The Intermediate Value Theorem implies that there exists $p \in (a, b)$ for which $h(p) = 0$. This number p is a fixed point for g because

$$0 = h(p) = g(p) - p \quad \text{implies that} \quad g(p) = p.$$

- (ii) Suppose, in addition, that $|g'(x)| \leq k < 1$ and that p and q are both fixed points in $[a, b]$. If $p \neq q$, then the Mean Value Theorem implies that a number ξ exists between p and q , and hence in $[a, b]$, with

$$\frac{g(p) - g(q)}{p - q} = g'(\xi).$$

Thus

$$|p - q| = |g(p) - g(q)| = |g'(\xi)| |p - q| \leq k |p - q| < |p - q|,$$

which is a contradiction. This contradiction must come from the only supposition, $p \neq q$. Hence, $p = q$ and the fixed point in $[a, b]$ is unique. ■ ■ ■

About the Order

- The order of fixed point iteration depends on $f(x)$
- Remember that $x_{n+1} = g(x_n)$
- If $|g'(r)| < 1 \rightarrow$ Converges
- If $g'(r) = 0 \rightarrow$ Converges Quadratically (at least)
- And if $g''(r) = 0 \rightarrow$ Converges Order 3 (at least)
- And so on.

Example 2 Show that $g(x) = (x^2 - 1)/3$ has a unique fixed point on the interval $[-1, 1]$.

Solution The maximum and minimum values of $g(x)$ for x in $[-1, 1]$ must occur either when x is an endpoint of the interval or when the derivative is 0. Since $g'(x) = 2x/3$, the function g is continuous and $g'(x)$ exists on $[-1, 1]$. The maximum and minimum values of $g(x)$ occur at $x = -1$, $x = 0$, or $x = 1$. But $g(-1) = 0$, $g(1) = 0$, and $g(0) = -1/3$, so an absolute maximum for $g(x)$ on $[-1, 1]$ occurs at $x = -1$ and $x = 1$, and an absolute minimum at $x = 0$.

Moreover

$$|g'(x)| = \left| \frac{2x}{3} \right| \leq \frac{2}{3}, \quad \text{for all } x \in (-1, 1).$$

So g satisfies all the hypotheses of Theorem 2.3 and has a unique fixed point in $[-1, 1]$. ■

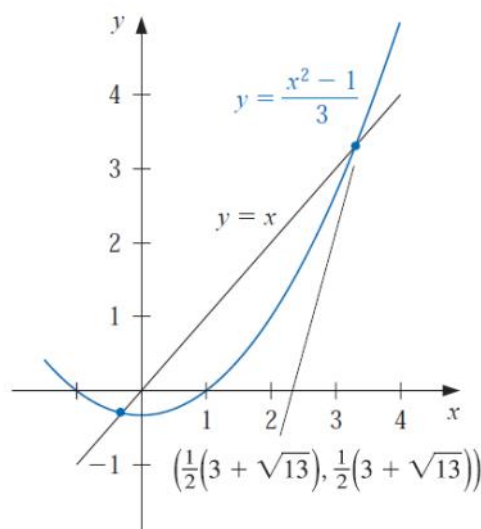
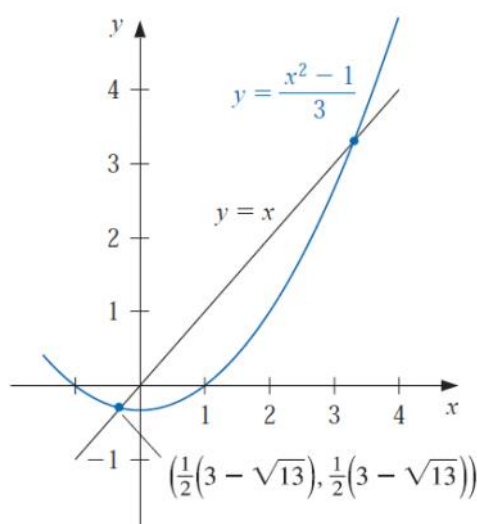
For the function in Example 2, the unique fixed point p in the interval $[-1, 1]$ can be determined algebraically. If

$$p = g(p) = \frac{p^2 - 1}{3}, \quad \text{then} \quad p^2 - 3p - 1 = 0,$$

which, by the quadratic formula, implies, as shown on the left graph in Figure 2.4, that

$$p = \frac{1}{2}(3 - \sqrt{13}).$$

Note that g also has a unique fixed point $p = \frac{1}{2}(3 + \sqrt{13})$ for the interval $[3, 4]$. However, $g(4) = 5$ and $g'(4) = \frac{8}{3} > 1$, so g does not satisfy the hypotheses of Theorem 2.3 on $[3, 4]$. This demonstrates that the hypotheses of Theorem 2.3 are sufficient to guarantee a unique fixed point but are not necessary. (See the graph on the right in Figure 2.5.)



Example 3 Show that Theorem 2.3 does not ensure a unique fixed point of $g(x) = 3^{-x}$ on the interval $[0, 1]$, even though a unique fixed point on this interval does exist.

Solution $g'(x) = -3^{-x} \ln 3 < 0$ on $[0, 1]$, the function g is strictly decreasing on $[0, 1]$. So

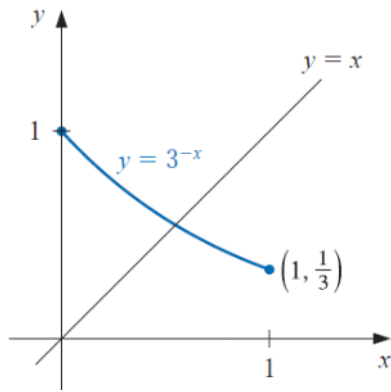
$$g(1) = \frac{1}{3} \leq g(x) \leq 1 = g(0), \quad \text{for} \quad 0 \leq x \leq 1.$$

Thus, for $x \in [0, 1]$, we have $g(x) \in [0, 1]$. The first part of Theorem 2.3 ensures that there is at least one fixed point in $[0, 1]$.

However,

$$g'(0) = -\ln 3 = -1.098612289,$$

so $|g'(x)| \not\leq 1$ on $(0, 1)$, and Theorem 2.3 cannot be used to determine uniqueness. But g is always decreasing, and it is clear from Figure 2.6 that the fixed point must be unique. ■



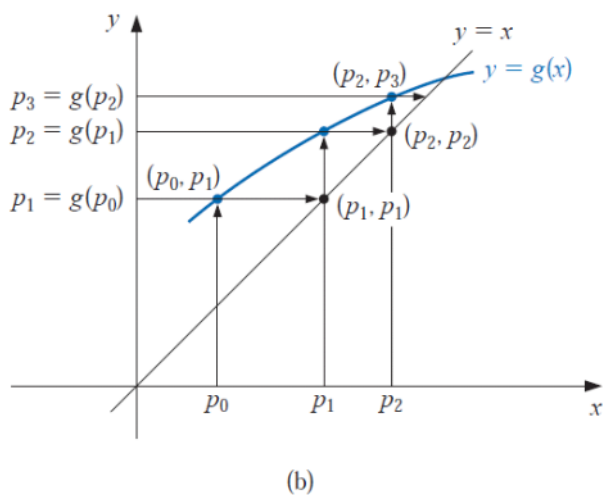
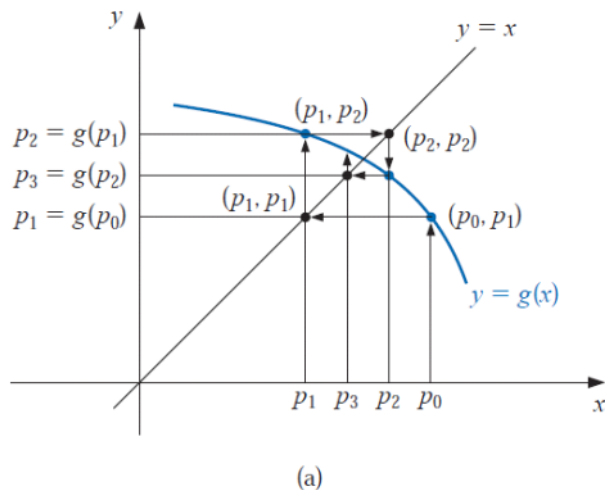
Fixed-Point Iteration

We cannot explicitly determine the fixed point in Example 3 because we have no way to solve for p in the equation $p = g(p) = 3^{-p}$. We can, however, determine approximations to this fixed point to any specified degree of accuracy. We will now consider how this can be done.

To approximate the fixed point of a function g , we choose an initial approximation p_0 and generate the sequence $\{p_n\}_{n=0}^{\infty}$ by letting $p_n = g(p_{n-1})$, for each $n \geq 1$. If the sequence converges to p and g is continuous, then

$$p = \lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} g(p_{n-1}) = g\left(\lim_{n \rightarrow \infty} p_{n-1}\right) = g(p),$$

and a solution to $x = g(x)$ is obtained. This technique is called **fixed-point**, or **functional iteration**. The procedure is illustrated in Figure 2.7 and detailed in Algorithm 2.2.



Fixed-Point Iteration

To find a solution to $p = g(p)$ given an initial approximation p_0 :

INPUT initial approximation p_0 ; tolerance TOL ; maximum number of iterations N_0 .

OUTPUT approximate solution p or message of failure.

Step 1 Set $i = 1$.

Step 2 While $i \leq N_0$ do Steps 3–6.

Step 3 Set $p = g(p_0)$. (Compute p_i .)

Step 4 If $|p - p_0| < TOL$ then
 OUTPUT (p); (The procedure was successful.)
 STOP.

Step 5 Set $i = i + 1$.

Step 6 Set $p_0 = p$. (Update p_0 .)

Step 7 OUTPUT ('The method failed after N_0 iterations, $N_0 =$ ', N_0);
 (The procedure was unsuccessful.)
 STOP.

The equation $x^3 + 4x^2 - 10 = 0$ has a unique root in $[1, 2]$. There are many ways to change the equation to the fixed-point form $x = g(x)$ using simple algebraic manipulation. For example, to obtain the function g described in part (c), we can manipulate the equation $x^3 + 4x^2 - 10 = 0$ as follows:

$$4x^2 = 10 - x^3, \quad \text{so} \quad x^2 = \frac{1}{4}(10 - x^3), \quad \text{and} \quad x = \pm \frac{1}{2}(10 - x^3)^{1/2}.$$

To obtain a positive solution, $g_3(x)$ is chosen. It is not important for you to derive the functions shown here, but you should verify that the fixed point of each is actually a solution to the original equation, $x^3 + 4x^2 - 10 = 0$.

(a) $x = g_1(x) = x - x^3 - 4x^2 + 10$

(b) $x = g_2(x) = \left(\frac{10}{x} - 4x\right)^{1/2}$

(c) $x = g_3(x) = \frac{1}{2}(10 - x^3)^{1/2}$

(d) $x = g_4(x) = \left(\frac{10}{4 + x}\right)^{1/2}$

(e) $x = g_5(x) = x - \frac{x^3 + 4x^2 - 10}{3x^2 + 8x}$

n	(a)	(b)	(c)	(d)	(e)
0	1.5	1.5	1.5	1.5	1.5
1	-0.875	0.8165	1.286953768	1.348399725	1.373333333
2	6.732	2.9969	1.402540804	1.367376372	1.365262015
3	-469.7	$(-8.65)^{1/2}$	1.345458374	1.364957015	1.365230014
4	1.03×10^8		1.375170253	1.365264748	1.365230013
5			1.360094193	1.365225594	
6			1.367846968	1.365230576	
7			1.363887004	1.365229942	
8			1.365916734	1.365230022	
9			1.364878217	1.365230012	
10			1.365410062	1.365230014	
15			1.365223680	1.365230013	
20			1.365230236		
25			1.365230006		
30			1.365230013		

The actual root is 1.365230013, as was noted in Example 1 of Section 2.1. Comparing the results to the Bisection Algorithm given in that example, it can be seen that excellent results have been obtained for choices (c), (d), and (e) (the Bisection method requires 27 iterations for this accuracy). It is interesting to note that choice (a) was divergent and that (b) became undefined because it involved the square root of a negative number. $\square$

① Use a fixed-point iteration method to determine a solution for $x^3 + x - 1 = 0$ on $[1, 2]$. Use $x_0 = 1.5$ and perform exactly 4 iterations.

Theorem 2.4 (Fixed-Point Theorem)

Let $g \in C[a, b]$ be such that $g(x) \in [a, b]$, for all x in $[a, b]$. Suppose, in addition, that g' exists on (a, b) and that a constant $0 < k < 1$ exists with

$$|g'(x)| \leq k, \quad \text{for all } x \in (a, b).$$

Then for any number p_0 in $[a, b]$, the sequence defined by

$$p_n = g(p_{n-1}), \quad n \geq 1,$$

converges to the unique fixed point p in $[a, b]$. $\blacksquare$

Proof Theorem 2.3 implies that a unique point p exists in $[a, b]$ with $g(p) = p$. Since g maps $[a, b]$ into itself, the sequence $\{p_n\}_{n=0}^{\infty}$ is defined for all $n \geq 0$, and $p_n \in [a, b]$ for all n . Using the fact that $|g'(x)| \leq k$ and the Mean Value Theorem 1.8, we have, for each n ,

$$|p_n - p| = |g(p_{n-1}) - g(p)| = |g'(\xi_n)| |p_{n-1} - p| \leq k |p_{n-1} - p|,$$

where $\xi_n \in (a, b)$. Applying this inequality inductively gives

$$|p_n - p| \leq k |p_{n-1} - p| \leq k^2 |p_{n-2} - p| \leq \cdots \leq k^n |p_0 - p|. \quad (2.4)$$

Since $0 < k < 1$, we have $\lim_{n \rightarrow \infty} k^n = 0$ and

$$\lim_{n \rightarrow \infty} |p_n - p| \leq \lim_{n \rightarrow \infty} k^n |p_0 - p| = 0.$$

Hence $\{p_n\}_{n=0}^{\infty}$ converges to p . $\blacksquare \blacksquare \blacksquare$

Corollary 2.5 If g satisfies the hypotheses of Theorem 2.4, then bounds for the error involved in using p_n to approximate p are given by

$$|p_n - p| \leq k^n \max\{p_0 - a, b - p_0\} \quad (2.5)$$

and

$$|p_n - p| \leq \frac{k^n}{1 - k} |p_1 - p_0|, \quad \text{for all } n \geq 1. \quad (2.6)$$

Both inequalities in the corollary relate the rate at which $\{p_n\}_{n=0}^{\infty}$ converges to the bound k on the first derivative. The rate of convergence depends on the factor k^n . The smaller the value of k , the faster the convergence, which may be very slow if k is close to 1. $\blacksquare$

Illustration Let us reconsider the various fixed-point schemes described in the preceding illustration in light of the Fixed-point Theorem 2.4 and its Corollary 2.5.

- (a) For $g_1(x) = x - x^3 - 4x^2 + 10$, we have $g_1(1) = 6$ and $g_1(2) = -12$, so g_1 does not map $[1, 2]$ into itself. Moreover, $g'_1(x) = 1 - 3x^2 - 8x$, so $|g'_1(x)| > 1$ for all x in $[1, 2]$. Although Theorem 2.4 does not guarantee that the method must fail for this choice of g , there is no reason to expect convergence.
- (b) With $g_2(x) = [(10/x) - 4x]^{1/2}$, we can see that g_2 does not map $[1, 2]$ into $[1, 2]$, and the sequence $\{p_n\}_{n=0}^{\infty}$ is not defined when $p_0 = 1.5$. Moreover, there is no interval containing $p \approx 1.365$ such that $|g'_2(x)| < 1$, because $|g'_2(p)| \approx 3.4$. There is no reason to expect that this method will converge.
- (c) For the function $g_3(x) = \frac{1}{2}(10 - x^3)^{1/2}$, we have

$$g'_3(x) = -\frac{3}{4}x^2(10 - x^3)^{-1/2} < 0 \quad \text{on } [1, 2],$$

so g_3 is strictly decreasing on $[1, 2]$. However, $|g'_3(2)| \approx 2.12$, so the condition $|g'_3(x)| \leq k < 1$ fails on $[1, 2]$. A closer examination of the sequence $\{p_n\}_{n=0}^{\infty}$ starting with $p_0 = 1.5$ shows that it suffices to consider the interval $[1, 1.5]$ instead of $[1, 2]$. On this interval it is still true that $g'_3(x) < 0$ and g_3 is strictly decreasing, but, additionally,

$$1 < 1.28 \approx g_3(1.5) \leq g_3(x) \leq g_3(1) = 1.5,$$

for all $x \in [1, 1.5]$. This shows that g_3 maps the interval $[1, 1.5]$ into itself. It is also true that $|g'_3(x)| \leq |g'_3(1.5)| \approx 0.66$ on this interval, so Theorem 2.4 confirms the convergence of which we were already aware.

- (d) For $g_4(x) = (10/(4 + x))^{1/2}$, we have

$$|g'_4(x)| = \left| \frac{-5}{\sqrt{10}(4 + x)^{3/2}} \right| \leq \frac{5}{\sqrt{10}(5)^{3/2}} < 0.15, \quad \text{for all } x \in [1, 2].$$

The bound on the magnitude of $g'_4(x)$ is much smaller than the bound (found in (c)) on the magnitude of $g'_3(x)$, which explains the more rapid convergence using g_4 .

- (e) The sequence defined by

$$g_5(x) = x - \frac{x^3 + 4x^2 - 10}{3x^2 + 8x}$$

converges much more rapidly than our other choices. In the next sections we will see where this choice came from and why it is so effective. □

From what we have seen.

Although the various functions we have given are fixed-point problems for the same root-finding problem, they differ vastly as techniques for approximating the solution to the root-finding problem. Their purpose is to illustrate what needs to be answered:

- **Question:** How can we find a fixed-point problem that produces a sequence that reliably and rapidly converges to a solution to a given root-finding problem?

The following theorem and its corollary give us some clues concerning the paths we should pursue and, perhaps more importantly, some we should reject.

1. Use algebraic manipulation to show that each of the following functions has a fixed point at p precisely when $f(p) = 0$, where $f(x) = x^4 + 2x^2 - x - 3$.
- a. $g_1(x) = (3 + x - 2x^2)^{1/4}$ b. $g_2(x) = \left(\frac{x + 3 - x^4}{2}\right)^{1/2}$
- c. $g_3(x) = \left(\frac{x + 3}{x^2 + 2}\right)^{1/2}$ d. $g_4(x) = \frac{3x^4 + 2x^2 + 3}{4x^3 + 4x - 1}$
2. a. Perform four iterations, if possible, on each of the functions g defined in Exercise 1. Let $p_0 = 1$ and $p_{n+1} = g(p_n)$, for $n = 0, 1, 2, 3$.
- b. Which function do you think gives the best approximation to the solution?